THE CURIOSITY CUP 2026
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Socially Optimal Pricing and Capacity Dynamics in a Competitive EV Charging Corridor

ORSOL Optimization Team

Nakul Sreekanth, Bhavya Lakhina, Zephan Thomas

Purdue University

ABSTRACT

Public Electric Vehicle (EV) charging networks are essential to sustainable transportation, yet the pricing and capacity dynamics of competitive highway corridors remain poorly understood. This study develops a SAS-based framework to evaluate how station-level pricing along California's I-210 corridor affects not only profits, but also system efficiency, congestion, and equitable access to clean transportation. Using Caltrans PeMS traffic volumes, CAISO wholesale electricity prices, and detailed station metadata, we estimate hourly charging demand through a logit discrete-choice model with a home-charging outside option and compute a Nash-like equilibrium using an iterative best-response algorithm in PROC OPTMODEL. Results show that the corridor behaves as a highly competitive market: prices remain stable even during peak demand, capacity constraints-not pricing power-drive utilization, and profits are evenly distributed across stations. These findings highlight how competitive EV charging markets can support affordability, reduce congestion, and guide socially optimal infrastructure planning as EV adoption accelerates.

INTRODUCTION

As electric vehicle adoption accelerates, public charging infrastructure has become a critical component of sustainable transportation systems, influencing emissions, congestion, and equitable access to clean mobility. Highway corridors are especially important because drivers have limited flexibility in choosing where to charge, yet stations often cluster near major exits, creating competitive environments that shape both pricing and system performance. California's I-210 corridor-home to more than fifty fast-charging stations drawing from the same stream of freeway traffic-offers a natural setting to study how competition, capacity constraints, and driver behavior interact. Understanding these dynamics is essential for identifying socially optimal outcomes, such as maintaining affordability, reducing bottlenecks, and ensuring efficient use of energy resources. This paper develops a SAS-based modeling framework that integrates traffic data, electricity prices, and station characteristics to estimate hourly demand and compute equilibrium prices, providing insights into how competitive charging networks can support sustainable, corridor-scale EV infrastructure planning.

DATA

Our analysis integrates several publicly available datasets describing traffic conditions, electricity prices, station characteristics, and EV charging behavior along the I-210 corridor. Hourly traffic volumes were obtained from the Caltrans Performance Measurement System (PeMS), which provides sensor-level counts for freeway mainline and offramp locations. This data allows us to estimate the pool of EVs traveling along the corridor and the subset likely to require charging each hour. Wholesale electricity prices were sourced from the California

Independent System Operator (CAISO) real-time market and converted from dollars per megawatt-hour to dollars per kilowatt-hour to represent marginal energy costs.

Station metadata were collected from OpenChargeMap, including geographic coordinates, charger power ratings, connector types, and port counts. These attributes determine each station's effective capacity and influence the distance component of the driver choice model. To approximate typical charging behavior, we incorporated Distributed Electrical Systems Laboratory (DESL) charging-session data, which provide empirical distributions of session duration and energy delivered per session, allowing us to validate and refine industry-standard behavioral assumptions. All datasets were later aligned to a common hourly time index and prepared for analysis.

PROBLEM

We study competitive pricing behavior among EV charging stations along California's I-210 corridor, where more than fifty fast-charging stations draw from the same stream of freeway traffic, face similar wholesale electricity costs, and operate under physical capacity constraints. Drivers choose among stations based on price and detour distance, with the option to forgo public charging and rely on home charging instead. The outside option is anchored to an average residential electricity cost of $0.18 per kWh, consistent with recent national estimates (Kurczewski, 2024), providing a realistic benchmark for opting out of corridor charging.

Because each station's optimal price depends on the prices set by its competitors, the corridor forms a strategic environment. We model this setting as a Nash-like equilibrium in which each station maximizes hourly profit subject to capacity limits and demand derived from a logit discrete-choice model. The challenge is to integrate heterogeneous datasets, estimate demand, and compute equilibrium prices in a way that reflects real-world operational constraints. SAS provides the tools needed to perform this integration, solve the nonlinear optimization problem, and analyze the resulting market dynamics.

DATA CLEANING/VALIDATION

Our data pipeline integrates five public sources: Caltrans PeMS traffic sensors, CAISO wholesale electricity prices, OpenChargeMap station metadata and DESL charging-session data. Because these sources differ in structure, units, and granularity, we implemented a standardized cleaning, optimization and visualization workflow in **SAS Viya Workbench for Learners®** to produce a unified hourly dataset for modeling.

CAISO electricity prices were retrieved through the OASIS API, which reports values in dollars per megawatt-hour with occasional duplicate hourly entries. Using **PROC IMPORT** and **PROC SORT**, we converted prices to dollars per kilowatt-hour, corrected hour-24 timestamps to hour-0, and removed duplicate observations. This produced 8,700 unique hourly price records aligned to a consistent 0–23 hour index.

PeMS traffic data were obtained from 56 freeway exit sensors, each returning 24 hourly counts in text format. After importing these files with **PROC IMPORT**, we standardized column names and validated numeric fields. Using **PROC MEANS**, we aggregated traffic across all sensors to create a single corridor-level hourly demand series, reducing 1,344 sensor-hour records to 24 hourly totals that reflect the shared traffic stream to all stations.

Station metadata from OpenChargeMap were flattened from JSON into tabular form and cleaned by harmonizing coordinate formats, converting charger power ratings to kilowatts, and removing stations with incomplete location information. This produced a consistent dataset of 51 stations with standardized capacity attributes.

Charging-session behavior was derived from 10,247 DESL records, which included special characters in column names and energy delivered in watt-hours. Using **PROC SQL**, we

renamed escaped variables, converted energy to kilowatt-hours, and computed hourly averages for session duration and energy delivered, yielding 24 hourly behavioral parameters used to scale demand.

Across all datasets, missing or anomalous values were identified using **PROC MEANS** and **PROC FREQ**. Fewer than 0.1% of CAISO price records were missing; these were imputed using a conservative baseline value to avoid biasing equilibrium prices upward. Finally, all cleaned datasets were merged on a common hourly timestamp using **PROC SQL**, and only the converged iteration of the optimization output was retained. The final merged dataset contained 1,224 complete station-hour observations with no join loss.

METHODOLOGY

Our methodology combines a discrete-choice demand model with a station-level profit-maximization framework to capture how drivers and charging stations interact along the I-210 corridor. Each driver evaluates stations based on two main factors: the price they would pay, and the detour distance required to reach the station. These factors together determine the "utility" or attractiveness of each station, which we formalize in Appendix Equation (1).

Using these utilities, we estimate the probability that a driver selects each station through a standard logit choice rule, shown in Appendix Equation (2). Expected hourly demand at each station is then obtained by aggregating these choice probabilities across all drivers, as defined in Appendix Equation (3). To calibrate how sensitive drivers are to price, we use published estimates of EV charging price elasticity, which report values as low as −1.5 in public fast-charging contexts (Zhang et al., 2024). These elasticity targets allow us to compute station-specific scale parameters and then take their corridor-wide average, as summarized in Appendix Equation (4).

Each station then chooses the price that maximizes its hourly profit, defined as the margin between its retail price and wholesale electricity cost multiplied by expected demand (Appendix Equation (5)). Because stations operate under physical capacity limits, we incorporate a smooth penalty term in the profit function that discourages prices leading to overload conditions; the resulting optimization problem is presented in Appendix Equation (5). We solve this nonlinear problem in **PROC OPTMODEL**. Since stations interact strategically, we compute a Nash-like equilibrium using an iterative best-response algorithm with a damping factor, updating prices until changes fall below a predefined threshold and yielding a stable set of hourly equilibrium prices for the entire corridor.

ANALYSIS

The equilibrium pricing model provides a detailed view of how competition shapes charging behavior along the I-210 corridor. Figure 1 presents a dual-axis plot created in **PROC SGPLOT**, showing hourly demand, the average equilibrium price, and the price trajectories of three representative stations. Demand peaks sharply around hour 18, reflecting the evening surge in freeway traffic. Yet prices remain remarkably stable throughout the day, with no corresponding spike during the demand peak. This stability-despite a several-fold increase in demand-highlights the strength of competitive pressure: stations cannot

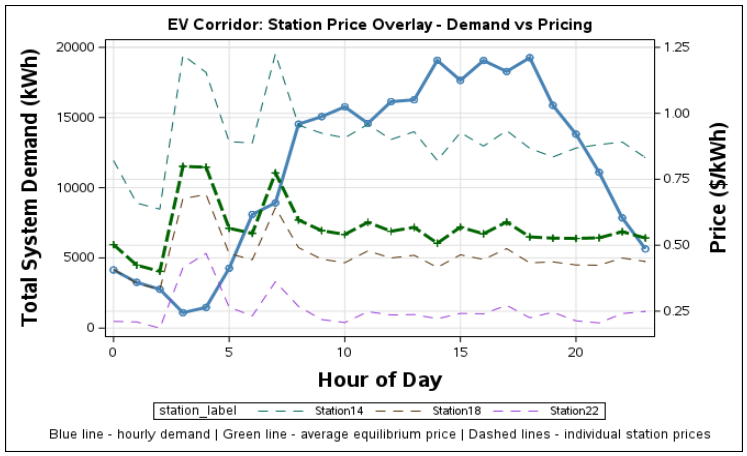

Figure 1

raise prices without losing customers to nearby alternatives, helping maintain affordability for drivers.

Capacity utilization follows a different dynamic. Figure 2 displays a heatmap generated in **PROC SGPLOT** using a **custom colormodel gradient**. Stations operate well below capacity during hours 0–6, but utilization rises sharply beginning around hour 6 and remains extremely elevated through hour 23. In many afternoon and evening hours, stations exceed their rated capacity by more than 200–300%, indicating that demand far outstrips the physical infrastructure available. This severe over-capacity reinforces that physical constraints- not pricing power- are the dominant factor shaping station performance during high-traffic periods. The pattern suggests that expanding charging capacity would be substantially more effective than price adjustments for reducing congestion, improving reliability, and supporting sustainable corridor operations

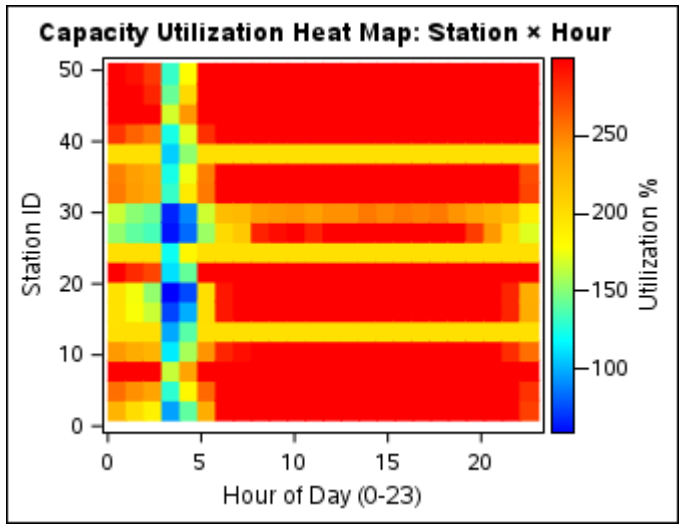

Figure 2

Profit distribution across stations is similarly balanced. Figure 3 shows a mid-day snapshot (hour 12) of profit shares across the network, produced using **PROC GCHART**. Profits are spread evenly, with no station capturing a dominant share. This even distribution persists throughout the day, and an animation included as Appendix Video A1 illustrates how profit concentration decreases slightly during high-demand hours as competition intensifies. Together, these results show that the corridor behaves much like a perfectly competitive market, with limited opportunities for stations to exercise pricing power and strong incentives to maintain efficient, user-friendly service.

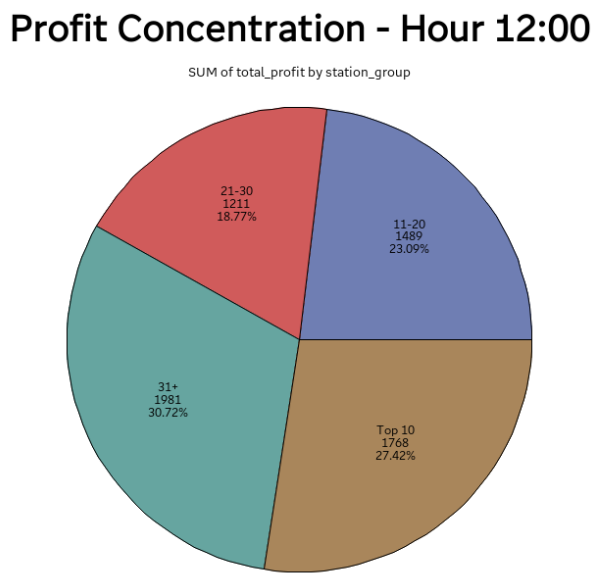

Figure 3

Appendix Figure A1, created in **SAS Visual Analytics**, provides a geospatial map of the corridor, showing the locations of charging stations relative to freeway exits and traffic flows. Appendix Figure A2 presents the convergence behavior of the iterative best-response algorithm. The curve shows mild oscillation in early iterations followed by rapid stabilization, confirming that the equilibrium is both stable and computationally tractable within the SAS optimization environment.

DISCUSSION

The results reveal several defining features of competitive EV charging markets. First, equilibrium prices remain stable even during periods of high demand, indicating that competition-not scarcity-prevents price spikes that could undermine affordability and

discourage EV adoption. Second, consistently high utilization levels show that physical capacity, rather than pricing power, constrains station performance; many stations operate near their limits for much of the day, underscoring the need for targeted infrastructure expansion to reduce congestion and improve reliability. Third, profits are evenly distributed across operators, reflecting the corridor's dense and geographically balanced network and suggesting that competitive dynamics are robust even in high-traffic environments. Finally, the rapid convergence of the iterative best-response algorithm demonstrates that SAS can efficiently solve nonlinear, multi-agent optimization problems, providing a scalable framework for evaluating corridor-level charging strategies and supporting data-driven planning for sustainable transportation systems.

SUGGESTIONS FOR FUTURE STUDIES

This study highlights several opportunities for extending corridor-level EV charging analysis. One direction is to incorporate richer behavioral data-such as observed charging session logs, real-time occupancy, or telematics-which would allow for more precise calibration of the discrete-choice model and better capture driver heterogeneity. Modeling distinct user groups, such as commuters, long-distance travelers, or rideshare drivers, could reveal how different populations respond to price and distance, improving the social equity implications of corridor planning. Incorporating dynamic electricity tariffs or demand-response programs would enable analysis of how stations adjust prices under more volatile cost conditions and how such policies influence grid sustainability. Finally, extending the framework to multiple interconnected corridors would allow researchers to study regional competition, spillover effects, and the broader system-level impacts of EV charging infrastructure on congestion, emissions, and energy resilience.

CONCLUSION

This paper presents a SAS-based framework for analyzing competitive pricing and capacity dynamics among EV charging stations along California's I-210 corridor. By integrating traffic volumes, wholesale electricity prices, and station characteristics, we estimate hourly demand using a logit discrete-choice model and compute equilibrium prices through an iterative best-response algorithm. Despite large fluctuations in demand, prices remain stable throughout the day, reflecting strong competitive pressure across the corridor. Capacity constraints-not pricing power-shape utilization patterns and influence the distribution of profits across stations.

These findings have important implications for sustainable corridor planning. Because competition naturally limits price increases, expanding station capacity may be more effective than regulating prices for reducing congestion, improving reliability, and supporting equitable access to clean transportation. The modeling approach demonstrates how SAS can unify data integration, nonlinear optimization, and visualization to inform socially optimal EV infrastructure decisions. As EV adoption accelerates, such analyses can guide investments that ensure charging networks remain efficient, competitive, and supportive of broader environmental and societal goals.

REFERENCES

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APPENDIX

$$U_{i,s} = -\theta e_i p_s - \beta_{\text{VOT}}\, d_{i,s}$$

Appendix Equation (1)

$$P_{i,s} = \frac{\exp\,(U_{i,s})}{\sum_{r \in S \cup \{0\}} \exp\,(U_{i,r})}$$

Appendix Equation (2)

$$E_s(p) = \sum_{i=1}^{N} e_i\; P_{i,s}(p)$$

Appendix Equation (3)

$$\pi_s(p) = (p_s - c_t)\, E_s(p)$$

Appendix Equation (4)

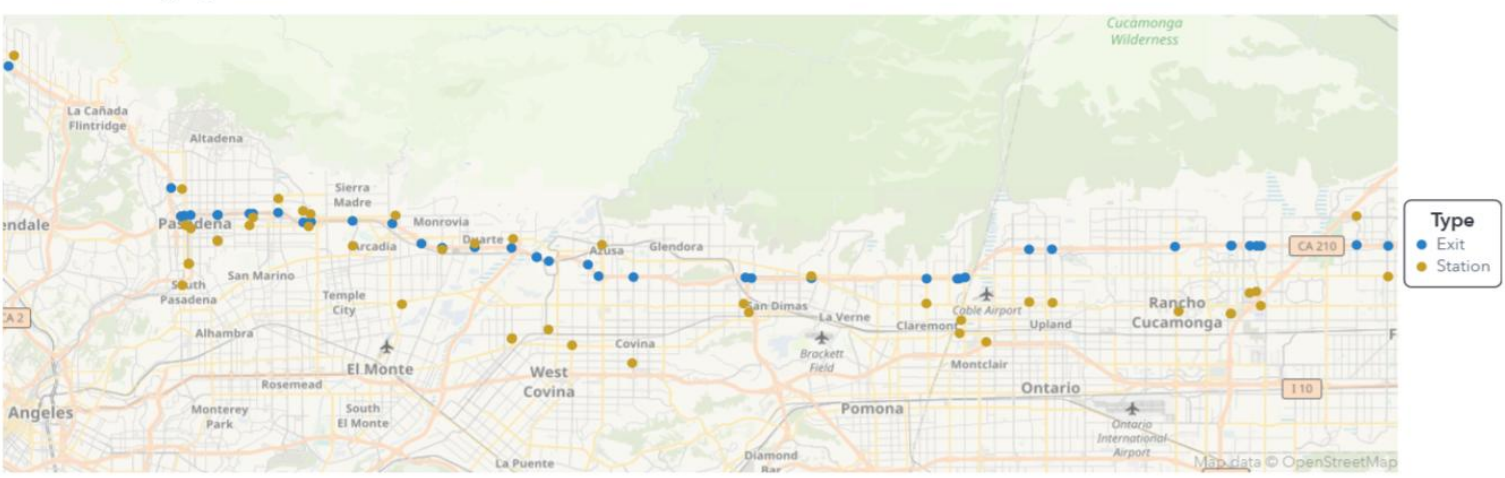
$$\max_{p_s}\;\; (p_s - c_t) \sum_{i=1}^{N} e_i\, P_{i,s}(p) + \gamma \left(\sum_{i=1}^{N} P_{i,s}\,(p)\mu - uN_s \right)$$

Appendix Equation (5)

Symbols and Definitions:

$U_{i,s}$— utility of station s for driver i
θ— price sensitivity parameter
e_i— energy required by driver i (kWh)
p_s— price charged by station s ($/kWh)
β_{VOT}— value-of-time coefficient (disutility per mile or minute)
$d_{i,s}$— detour distance for driver $i$to reach station s
$P_{i,s}$— probability that driver $i$chooses station
S— set of all public charging stations
0— outside option (home charging)
$E_s(p)$— expected hourly energy demand at station s
N— total number of drivers in the corridor
$\pi_s(p)$— hourly profit for station s
c_t— wholesale electricity cost at hour t($/kWh)
γ— penalty weight for exceeding capacity
μ— average energy delivered per charging session
u— utilization threshold (fraction of capacity)
N_s— number of charging ports at station s

Exit and Charging Stations on I-210 Districts 7 and 8



Appendix FigureA1

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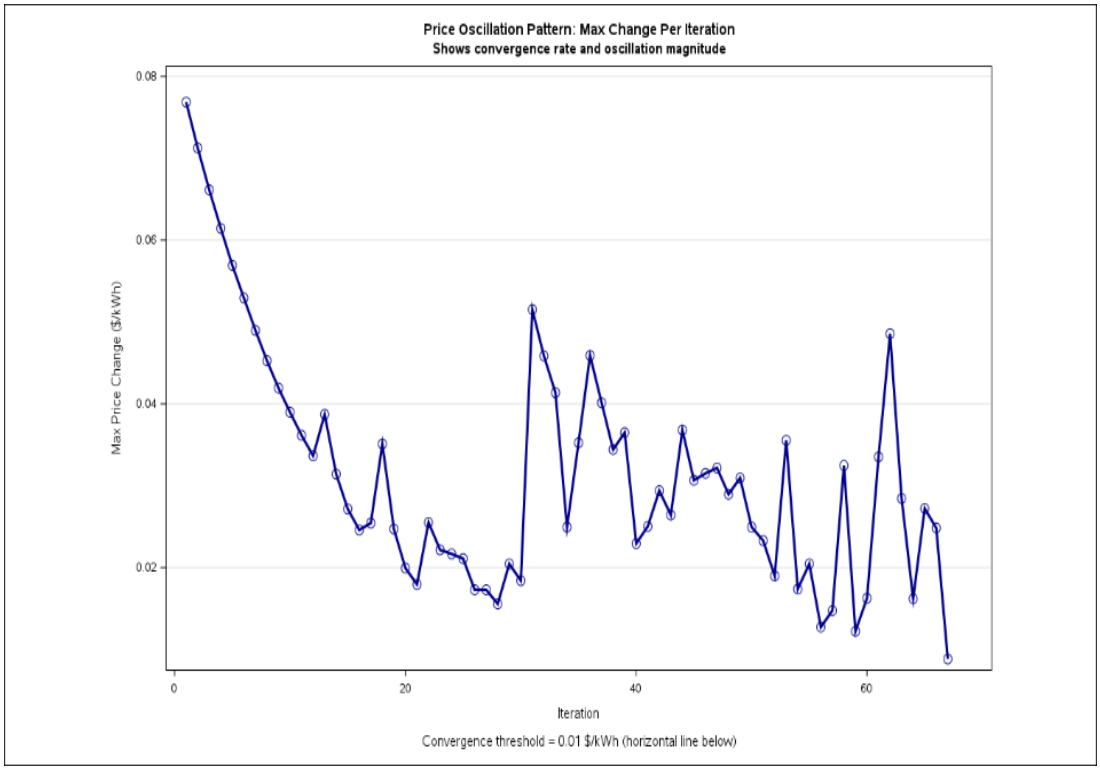

Appendix FigureA2

https://youtu.be/Zoz0iEQhMGk

Appendix VideoA1